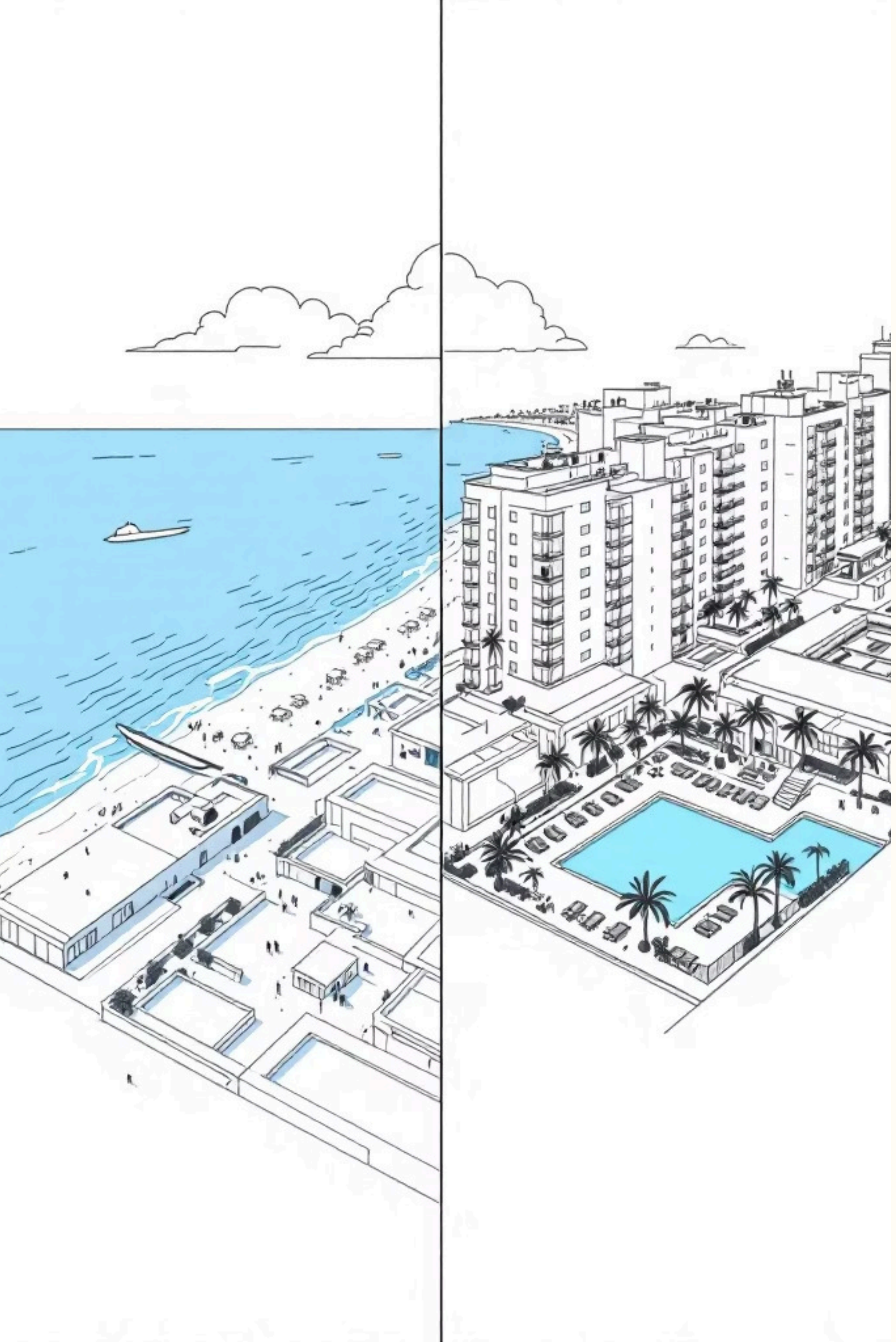


Hotel Booking Data Analysis & Insights

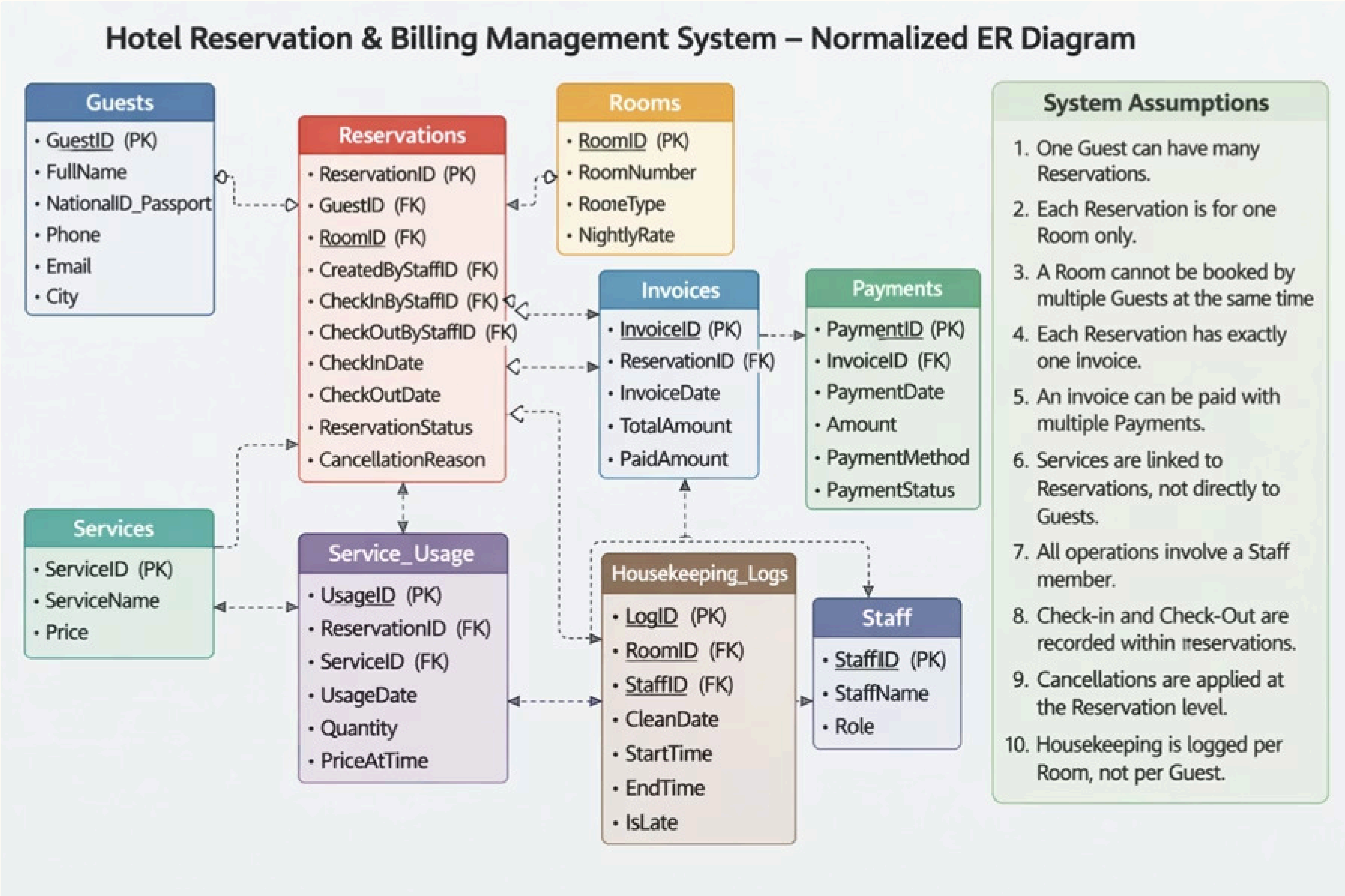
Data mining project using the Hotel Booking Demand dataset .

Objective: analyze booking behavior, identify risk drivers (cancellations, pricing, customer segments), and surface actionable recommendations for hotel operations and revenue teams.

HOTEL PROBLEMS:

1. Severe Cancellation: 87% of bookings have "No Deposit,"
2. their Corporate and Group segments are "price-locked"
3. Specific room types (B, C, F, G) and meal plans are rarely booked, representing wasted operational overhead.
4. They have (170+) minor countries with negligible volume, which clutters their market reporting and dilutes our ROI focus.

Database Schema



Summary of the 10 Analytical Deliverables

Guest Master View Displays the top 20 guests by lifetime spending and number of completed stays.

Purpose: Identify high-value / VIP customers for loyalty programs and targeted offers.

Key technique: Aggregating invoice totals only from checked-out reservations using LEFT JOINs.

Room Availability View Shows available vs booked rooms for any given date range.

Purpose: Enables real-time room inventory checks for front-desk staff.

Key technique: Non-overlapping interval check using NOT EXISTS.

Daily Occupancy Rate Calculates the percentage of occupied rooms for each day in a selected period.

Purpose: Monitor hotel fill rate and forecast demand.

Key technique: Date generation + LEFT JOIN with overlap condition Check-in ≤ date < Check-out.

Reservation Details View Comprehensive view of every reservation including guest name, room details, nights stayed, status, and room revenue.

Purpose: Quick reference for reservation audits and guest communication.

Key technique: Simple joins + DATEDIFF for nights calculation.

Cancellation Analysis Monthly cancellation rate with top 5 cancellation reasons.

Purpose: Understand revenue leakage patterns and improve booking policies.

Key technique: Conditional aggregation using CASE inside SUM/COUNT with FORMAT for monthly grouping.

Services Revenue View: Monthly revenue and usage count per service with top 5 services by revenue.

Purpose: Identify the most profitable ancillary services.

Key technique: Grouping by service name and month with multiplication for line revenue.

Invoice Aging & Outstanding Balances Classifies overdue invoices into aging buckets (0–7, 8–30, 31–60, 60+ days) and shows remaining balance.

Purpose: Support credit control and collection efforts.

Key technique: DATEDIFF combined with CASE statements for bucket classification.

Payment Method Breakdown Monthly breakdown of payments by method (Cash, Card, Online, etc.) including failed/refunded counts.

Purpose: Analyze customer payment preferences and payment gateway performance.

Key technique: Conditional summation per payment method.

Housekeeping & Room Turnover Report Per-room statistics including cleaning frequency, average cleaning time, and late cleaning percentage in the last 30 days.

Purpose: Evaluate housekeeping efficiency and identify operational bottlenecks.

Key technique: Time-difference calculation using DATEDIFF MINUTE and late flag aggregation.

Staff Performance View Measures staff contribution including reservations created, check-ins/outs processed, and revenue handled.

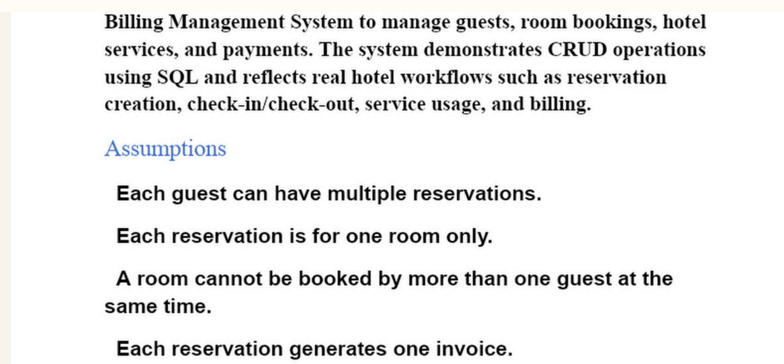
Purpose: Performance evaluation and incentive planning for reception and billing teams.

Key technique: Multiple conditional counts using CASE inside COUNT DISTINCT across staff role columns.

Google Docs

Hotel Reservation & Billing Management System

Hotel Reservation & Billing Management System Abstract This project aims to design ...and implement a Hotel Reservation and Billing Management System to manage


Billing Management System to manage guests, room bookings, hotel services, and payments. The system demonstrates CRUD operations using SQL and reflects real hotel workflows such as reservation creation, check-in/check-out, service usage, and billing.

Assumptions

Each guest can have multiple reservations.

Each reservation is for one room only.

A room cannot be booked by more than one guest at the same time.

Each reservation generates one invoice.

Data Cleaning & Preprocessing

Missing values

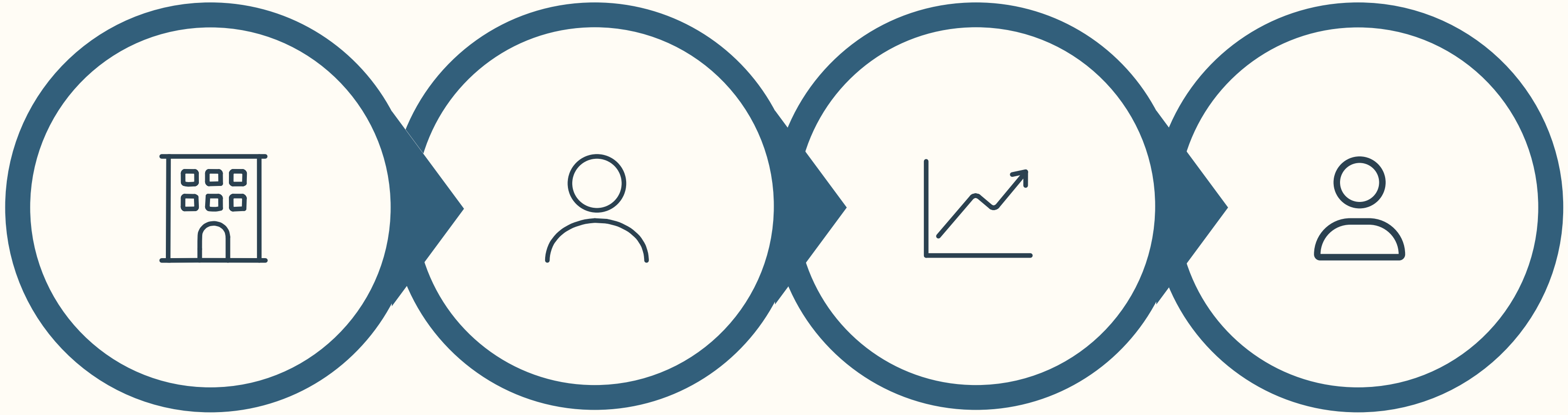
Handled by targeted imputation and row removal depending on field importance; dropped columns with excessive missingness (e.g., company).

Duplicates & validity

No duplicate bookings found; corrected invalid numeric values (negative ADR) and normalized date formats.

Feature engineering

Created lead_time bins, total_stay_nights, customer_type flags, and revenue_per_booking to support analysis and modeling.



Booking by Type

Customer Segments

ADR Distribution

Cancellation Correlations

This EDA sequence prioritized distributional checks first, then pricing behavior, and finally correlation checks to surface cancellation drivers and segment-level revenue patterns.

link: **https://github.com/Safaa9924/Hotel-project-Digilians-/blob/main/Python/hotel_pro7_checkpoint%20(1).ipynb**

OUR INSIGHT

```
l =====

    count   percen

l   79163        6
el  40046        3
```

Updated Distribution for: country

	Value	Count	Percentage (%)
0	PRT	48961	41.07
1	GBR	12119	10.17
2	FRA	10401	8.73
3	ESP	8560	7.18
4	DEU	7285	6.11
...	...	...	...
172	MRT	1	0.00
173	KIR	1	0.00
174	SDN	1	0.00
175	ATF	1	0.00
176	SLE	1	0.00

```
===== hotel =====

No rare categories (<3%)


===== arrival_date_month =====

No rare categories (<3%)


===== meal =====

Rare categories (<3%):
      count  percentage
meal
FB      798        0.67
```

Updated Distribution for: customer_segme

	Value	Count	Percentage (%)
0	Online TA	56408	47.32
1	Offline TA/TO	24182	20.29
2	Groups	19790	16.60
3	Direct	12582	10.55
4	Corporate	5282	4.43
5	Complementary	728	0.61
6	Aviation	235	0.20
7	Undefined	2	0.00

ted Distribution for: mea

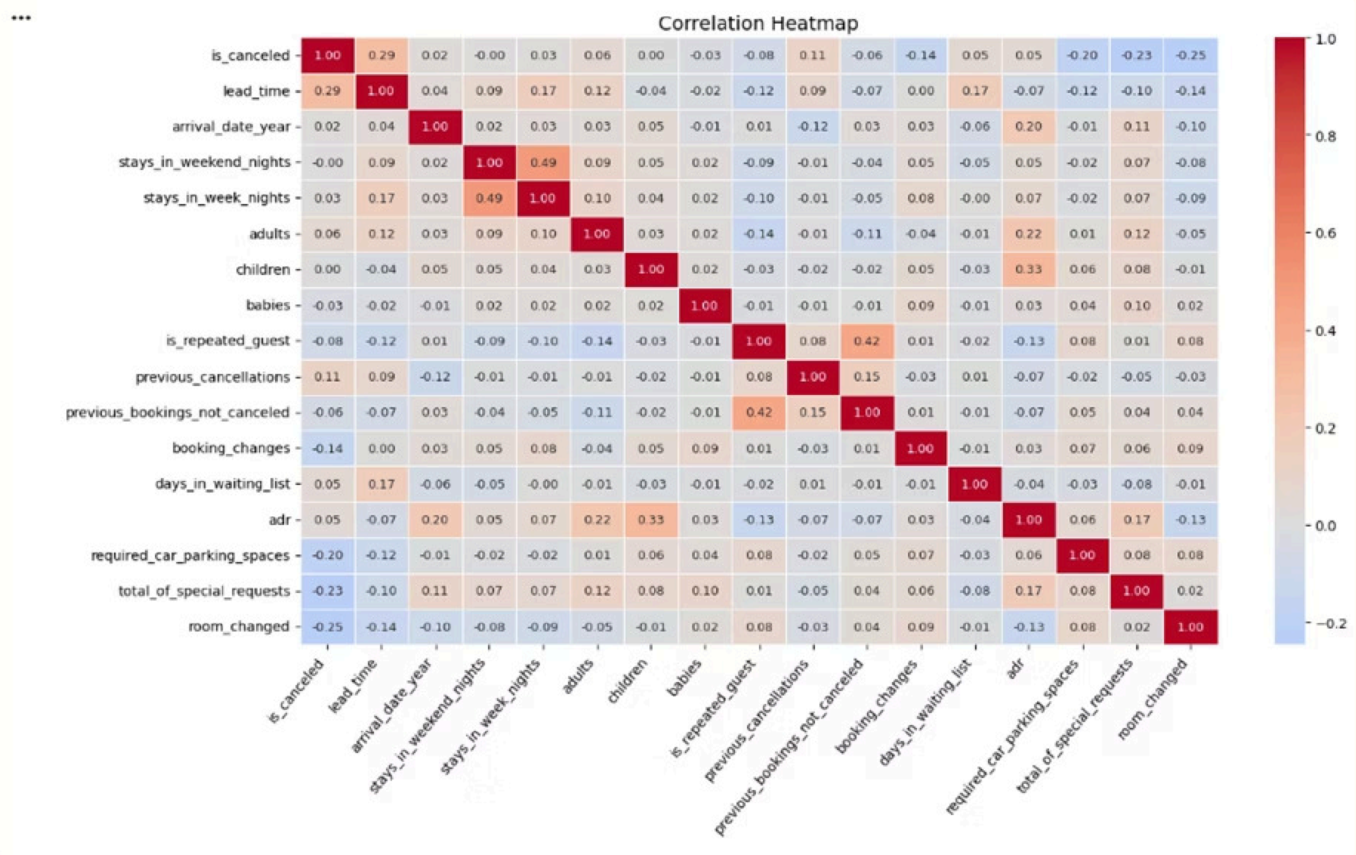
Value	Count	Percentage
BB	92235	77
HB	14458	12
SC	11718	9
FB	798	0

```
count   percentag

92235       77.3
14458       12.1
11718        9.8
  798        0.6
```

Booking & Customer Insights

- City Hotels dominate total bookings vs Resort Hotels (large urban demand concentration).
- Majority of bookings come through agents and tour operators — low direct-booking share.
- Customer mix: mix of transient, group, and contract bookings with clear revenue differences.

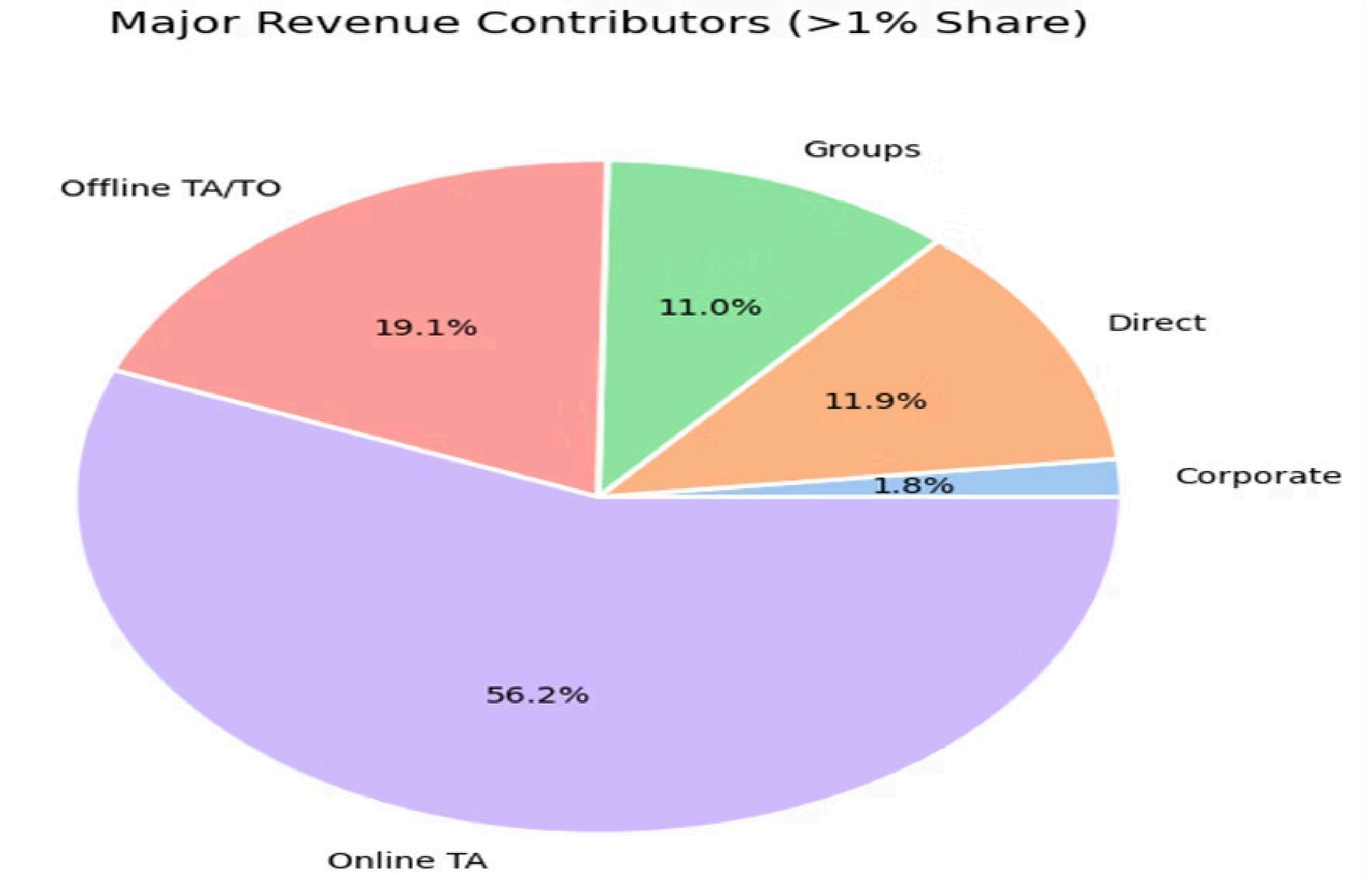


Revenue Dominance

Revenue Dominance: Online TA is the primary revenue driver at 56%, followed by Offline TA/TO at 19%. Small segments (Aviation, Corporate) were excluded from this view as their financial impact is negligible (near 0%).

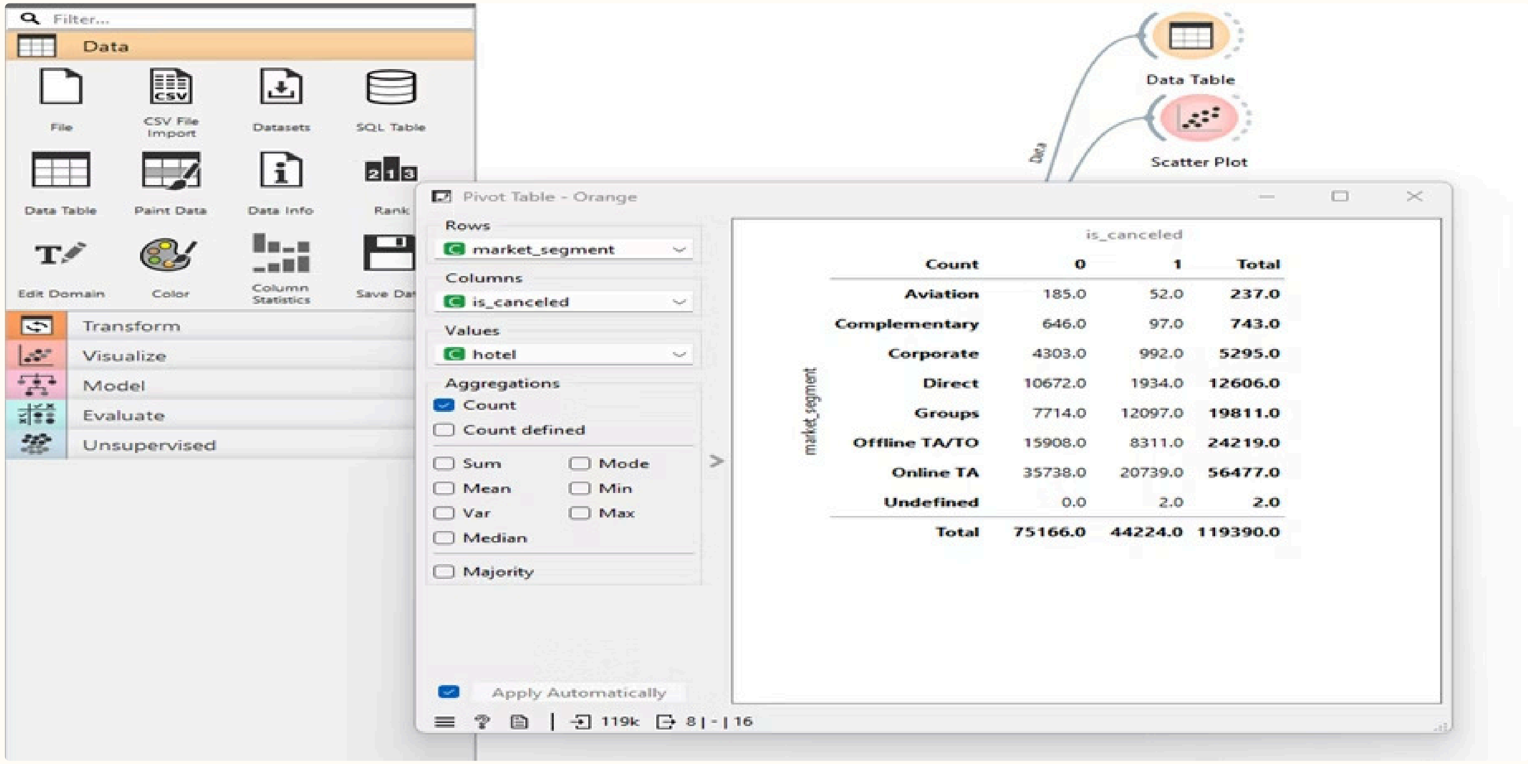
Action:

Since Online TA controls over half our revenue, negotiate better commission rates or shift 5-10% of this volume to Direct channels to save on fees.



Data Mining & Pattern Discovery

Model using (Orange)



Segmented guests into behavior clusters (short-stay business vs long-lead leisure) to tailor interventions.

Dashboard (Power BI) — KPIs & Interactivity

Core KPIs: total bookings, cancellations (37% overall), net revenue, ADR, and RevPAR. Interactive filters allow slicing by hotel type, market segment, arrival month, and distribution channel for operational decisions.

- Trend charts for cancellations vs bookings over time
- Segment comparison widgets (agent vs direct)
- Drill-through for booking-level detail

Representative SQL Patterns

1. JOIN bookings with customers and payments to compute revenue_per_booking and cancellation flags.
2. GROUP BY hotel, arrival_month to generate time-series KPIs.
3. Filter and window functions to identify repeated cancellers and average lead_time by segment.

Example: use windowed SUM() for rolling 30-day cancellation trends to trigger alerts.

Recommendations & Conclusion

1

1. Reduce cancellations

Introduce tiered deposit policies, targeted reminders for long-lead bookings, and gentle incentives for rebooking instead of refunds.

2

2. Boost direct bookings

Enhance direct-channel offers, loyalty perks, and personalized pricing to shift share away from agents where margins are lower.

3

3. Prioritize high-value segments

Allocate marketing and upsell capacity to segments that concentrate revenue; create family and corporate packages based on ADR patterns.

4

4. Embed analytics into ops

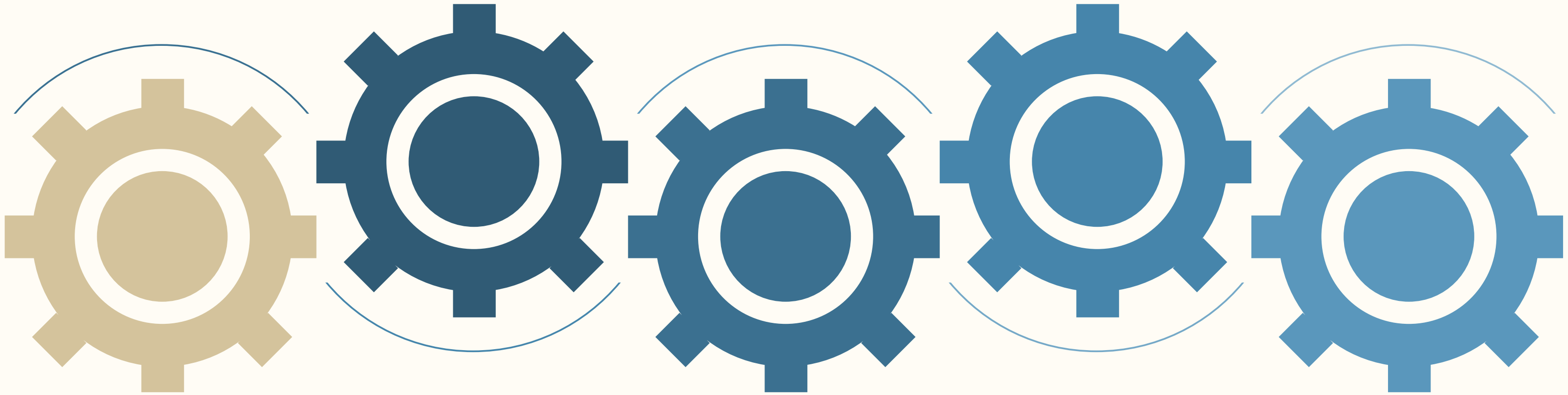
Operationalize predictive cancellation signals into booking workflows and front-desk alerts; monthly dashboard cadence for revenue ops.

Conclusion: This end-to-end analysis — from SQL extraction to Python/Orange modeling and Power BI dashboards — uncovers actionable levers to reduce cancellations, improve pricing, and grow profitable direct bookings. Next steps: pilot interventions for a representative property and measure impact.

Our solution:

1. Implement "Non-Refundable" Incentives: Offer a 5–10% discount on ADR (Average Daily Rate) for guests who choose a non-refundable "Prepaid" or "Non-Refundable" deposit type.
2. Instead of offering a fixed price (e.g., $100) all year round, you sign contracts with Corporate and Group clients based on a fixed percentage discount (e.g., 10% off) from the Best Available Rate (BAR) of the day
3. Inventory Consolidation: "Clean" the inventory by merging low-volume room types (B, C, F, G) into higher-performing categories through complimentary upgrades or renaming them for better market appeal.
4. Focus on Top-Tier Markets: Filter market reporting to focus exclusively on top-performing countries (e.g., PRT, GBR, FRA, ESP, DEU).

TEAM MEMBARES:



Shahinaz Abdelawaad

database creation

shahd najeh

insights sql

shaimaa ibrahim

EDA

shaimaa mohamed

EDA

safaa samy

PROBLEMS & SLOVING

THANKS